

RAVI PRAKASH

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WORK EXPERIENCE

Postdoctoral Associate 2026– Present
Mechanical Engineering
Johns Hopkins University, Baltimore, MD
With: Axel Krieger, Ph.D.

EDUCATION

Doctor of Philosophy 2022–2026
Mechanical Engineering and Materials Science
Duke University, Durham, NC
With: Patrick J. Codd, M.D. & Boyuan Chen, Ph.D.

Master of Science 2020–2021
Mechanical Engineering and Materials Science
Duke University, Durham, NC
With: Xiaoyue Ni, Ph.D.

Bachelor of Technology 2015–2019
Mechanical Engineering
National Institute of Technology, Warangal, India
With: P. Bangaru Babu, Ph.D.

SPECIALIZATION AND EXPERIENCE

Sensing for Informed Action

Postdoctoral associate specializing in **multimodal perception** and **closed-loop autonomy** for medical, surgical, and aerial robotic systems in complex, dynamic environments. I build **full-stack robotic systems** that connect **sensing**, **inference**, and **control** across acoustics, imaging, optomechanical systems, embedded platforms, and learning-based models to enable **reliable autonomy** in high-stakes, unstructured settings. My vision is to transform rich sensory data into **actionable intelligence** making precision robotic operation safer and more effective.

AWARDS

STAUBLI Medical Robotics Rising Star, Holistic Forum of Med. Rob. Junior Professors, ISMR	2026
Rhodes Graduate Fellowship for Interdisciplinary Research, Duke University	2025
IEEE-EMBS Body Sensor Network Travel Award	2025
Duke India Initiative Travel Award, Duke University	2023
Dean's Research Award for Master's Students	2021
Woo Center for Big Data and Precision Health Fellowship	2021
Design Health Fellow, Duke University	2021
Laboratory and Curriculum Development Fellowship, Duke University	2020
S.N. Bose Undergraduate Research Fellowship, Dept. Of Science and Technology, Government of India	2017
Merit Award in Physics and Information Technology, High School Diploma, Government of India	2015

PUBLICATIONS

* equal contribution ‡ equal advising; Updated 5th September, 2026

<i>Book Chapter [1]</i> • <i>Journal [3 Under Review / 3 Published]</i>
<i>Conference [10 Published]</i> • <i>Preprints/Workshops [2]</i>

Book Chapters

1. Chen, X., **Prakash, R.**, Jopling, J., Krieger, A. *Intelligent Robotic Assistants for Minimally Invasive Surgery: From Camera Holders to Autonomous Task Partners*.
Invited chapter for *AI in Robotic Surgery: Foundation, Innovation, and Future Direction*, Springer Nature.

Journal Articles

1. Liu, Y., **Prakash, R.**, Lo, L. Y., Roede, N., Chen, B. *Embodied Passive Aeroacoustic Perception Enables Relative Sensing and Pursuit Between Aerial Robots*.
Under review.
2. **Prakash, R.**, McNabb, R. P., Codd, P. J., Lin, S. *OCTN: Neural OCT Tissue Representations for Robot-Guided Precision Intervention*.
Under review.
3. Ma, G.*, **Prakash, R.***, Schleupner, B., Everitt, J. I., Mishra, A., Chen, J., Mann, B., Chen, B., Bridgeman, L., Zhong, P., Draelos, M., Eward, W., Codd, P. J. *TumorMap: A Laser-Based Surgical Platform for 3D Tumor Mapping and Fully-Automated Tumor Resection*.
Under review.
*** equal contribution.**
4. Chen, D., **Prakash, R.**, Wang, V., Chen, Z., Dias, S., Bridgeman, L., Oca, S., Mann, B. *Design and Evaluation of a Compliant Quasi Direct Drive End-Effector for Safe Robotic Ultrasound Imaging*.
IEEE Transactions on Medical Robotics and Bionics, 2026.
5. **Prakash, R.**, Dupre, M. E., Østbye, T., Xu, H. *Extracting Critical Information from Unstructured Clinicians' Notes Data to Identify Dementia Severity Using a Rule-Based Approach: Feasibility Study*.
Journal of Medical Internet Research (JMIR) Aging, 7, e57926, 2024.
6. Sperber, J., Zachem, T. J., **Prakash, R.**, Owolo, E., Yamamoto, K., Nguyen, A. D., Hockenberry, H., Ross, W. A., Herndon, J. E., Codd, P. J., Goodwin, C. R. *A Blinded Study Using Laser-Induced Endogenous Fluorescence Spectroscopy to Differentiate Ex Vivo Spine Tumor, Healthy Muscle, and Healthy Bone*.
Scientific Reports, 14(1), 1921, 2024.

Peer-Reviewed Conference Publications

Published

1. **Prakash, R.***, Wang, V.*, Mishra, A., Yuliarti, D., Zhong, P., McNabb, R., Codd, P. J.[‡], Bridgeman, L.[‡] *See, Plan, Cut: MPC-Based Autonomous Volumetric Robotic Laser Surgery with OCT Guidance*.
IEEE International Conference on Robotics and Automation (ICRA), 2026
*** equal contribution.**
2. Yuliarti, D.*, **Prakash, R.***, Cheung, H. H., Strong, A., Codd, P. J., Lin, S. *PalpAid: Multimodal Pneumatic Tactile Sensor for Tissue Palpation*.
IEEE 9th International Conference on Soft Robotics (RoboSoft), 2026, (pp. 660-665).
*** equal contribution.**
Mentored Project: Project conception, Experimental design, Paper writing.
3. Cheng, J., Zhao, X., Tripathi, S., Liu, S., Yu, X., **Prakash, R.**, Codd, P. J., Katz, J. E., Lin, S. *SurgXBench: Explainable Vision-Language Model Benchmark for Surgery*.
Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 8188-8198. 2026.
4. Chen, Z., Cahilig, A. C., Dias, S., Kolar, P., **Prakash, R.**, Codd, P. J. *Where is the Boundary? Multimodal Sensor Fusion Test Bench for Tissue Boundary Delineation*.
IEEE-EMBS International Conference on Body Sensor Networks (BSN), 2025.
Mentored Project: Student lead, Project conception, Paper writing.

5. Wang, V. Y., **Prakash, R.**, Oca, S. R., LoCicero, E. J., Codd, P. J., Bridgeman, L. J. *Sampling-Based Model Predictive Control for Volumetric Ablation in Robotic Laser Surgery*. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 4527–4533.
6. **Prakash, R.**, Wang, V., Chen, Z., Bridgeman, L., Codd, P. J. *Portable Dual Sensor Large Area Visualization System for Robotic Laser Surgery*. *International Symposium on Medical Robotics (ISMR)*, 2025, pp. 58–65.
7. **Prakash, R.**, Yamamoto, K. K., Oca, S. R., Ross, W., Codd, P. J. *Brain-Mimicking Phantom for Photoablation and Visualization*. *International Symposium on Medical Robotics (ISMR)*, 2023, pp. 1–7.
8. Ma, G., **Prakash, R.**, Mann, B., Ross, W., Codd, P. J. *3D Laser-and-Tissue Agnostic Data-Driven Method for Robotic Laser Surgical Planning*. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 8446–8453.
9. **Prakash, R.**, Dupre, M. E., Østbye, T., Xu, H. *A Rule-Based Framework to Identify Severity of Dementia from Unstructured Electronic Health Record Data*. *Conference Abstract: Presented at the Alzheimer’s Association International Conference (AAIC)*, 2023. *Journal Supplement: Published in Alzheimer’s & Dementia*, 19(S1): e075325, 2023.
10. Chatterjee, A., Valaparla, R. K., **Prakash, R.**, Balasubramanian, K. *Comparative Study of Fluid Flow and Heat Transfer in Microchannels with Uniformly Varying Cross-Section*. In *Proceedings of Emerging Trends in Mechanical Engineering*, 2019, pp. 25–30. Warangal, Telangana.

Preprints & Workshop Papers (Non–Peer-Reviewed)

1. Ross, W., **Prakash, R.**, Ma, G., Eward, W., Mann, B., Codd, P. J. *Optimization of Laser Photoablation for Robotic Soft-Tissue Surgery*. *Workshop on Data vs. Model in Medical Robotics (DMMR), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
2. Zachem, T. J., Chen, S. F., Venkatraman, V., Sykes, D. A., **Prakash, R.**, Spellicy, S., Suarez, A. D., Ross, W., Codd, P. J. *Computer Vision for Increased Operative Efficiency via Identification of Instruments in the Neurosurgical Operating Room: A Proof-of-Concept Study*. arXiv preprint, 2023.

CONFERENCE WORKSHOP ORGANIZED

1. Lin, S., **Prakash, R.**, Lu, J., Jiang, F., Dou, Q., Goldberg, K. *The Evolving Landscape of Surgical Robotics (ELSR): Foundation Models Meet New Robotic Embodiments*. *International Conference on Intelligent Robots & Systems (IROS)*, 2026, Pittsburgh, PA, United States.
2. Lin, S., **Prakash, R.**, Fichera, L., Mattos, L., Su, H., Yip, M. C., Codd, P., Webster, R. J. III. *The Evolving Landscape of Surgical Robotics (ELSR): Towards AI-Powered Unconventional Instruments and Technologies in the Operating Theatre*. *International Conference on Robotics and Automation (ICRA)*, 2025, Atlanta, Georgia, United States.

PATENTS

1. **Prakash, R.**, McNabb, R.P., Codd, P.J. SYSTEMS AND METHODS FOR NEURAL OCT TISSUE REPRESENTATIONS FOR ROBOT-GUIDED INTERVENTION. United States of America Patent Application No. 64/045,412. Provisional, Patent Pending
2. **Prakash, R.**, Chen, B., Codd, P.J., Lu, Yongjun. SYSTEMS AND METHODS FOR ACOUSTICALLY GUIDED SURGERY. United States of America Patent Application No. 63/988,766. Provisional, Patent Pending
3. Fudim, M., Richardson, E., Butch, J., Cross, V., Nandi, B., Lyon, R., **Prakash R.**, Ouyang, X., Miao, Y., Weil, B., Mitra, K., others. DEVICE FOR TREATING ORTHOSTATIC SYNCOPE AND METHODS THEREOF. United States patent application US 18/618,762. October 3, 2024.

INVITED TALKS

1. *Sensing-Driven Surgical Autonomy.*
Arizona State University, Medical Robotics Course, September 11, 2026 — Virtual.
2. *Multimodal Autonomous Robots for Healthcare.*
Johns Hopkins University, March 30, 2026 — Baltimore, Maryland, United States.
3. *Innovating Surgical Robotics: Enhancing Precision and Automation in Minimally Invasive Procedures.*
Indian Institute of Technology (IIT) Patna, February 9, 2024 — Patna, Bihar, India.
4. *Understanding the Basics of Natural Language Processing and Its Application to Processing Physicians' Notes.*
Duke Family Medicine and Community Health Grand Rounds, Duke University, September 14, 2021 — Durham, North Carolina, USA.

ABSTRACT PRESENTATIONS

Medical Robotics

1. **Prakash, R.***, Wang, V.*, Mishra, A., Yuliarti, D., Zhong, P., McNabb, R., Codd, P.J., Bridgeman, L. *See, Plan, Cut: MPC-Based Autonomous Volumetric Robotic Laser Surgery with OCT Guidance.*
[Winner] [Poster Presentation] Workshop on The Holistic Forum of Medical Robotics Junior Professors, International Symposium on Medical Robotics (ISMR), April 2026, Knoxville, TN, USA.
2. **Prakash, R.***, Wang, V.*, Gupta, K., Toshniwal M., Qian, S., Lu, Y., Rukhlin, A., McNabb, R., Codd, P., Bridgeman, L. *Design and Development of a Teleoperation Framework for a Robotic Laser Surgery Platform.*
[Poster Presentation] International Symposium on Medical Robotics (ISMR), April 2026, Knoxville, TN, USA.
3. Chen, Z.*, Cahilig, A. C.*, Dias, S., Kolar, P., **Prakash, R.**, Codd, P. J. *Where Is the Boundary: Multimodal Sensor Fusion Test Bench for Tissue Boundary Delineation.*
[Poster Presentation] Workshop on the Evolving Landscape of Surgical Robotics, IEEE International Conference on Robotics and Automation (ICRA), 2025, Atlanta, GA, USA.
4. **Prakash, R.**, Ma, G., Wang, V., Schleupner, B., Mishra, A., Everitt, J., Mann, B., Zhong, P., Bridgeman, L., Draelos, M., Chen, B., Eward, W., Codd, P. J. *Towards a Multimodal System for Non-Contact Robotic Surgery.*
[Poster Presentation] Duke Medical Robotics Symposium, October 25, 2024, Durham, NC, USA.
5. Chen, C., **Prakash, R.**, and Codd, P. J. *Precision in Practice: Enhancing Robotic Laser Surgery with Specialized Segmentation and Planning Web Application.*
[Poster Presentation] International Symposium on Medical Robotics (ISMR), April 2024, Atlanta, GA, USA.
6. Ma, G., **Prakash, R.**, Mann, B., Ross, W., Codd, P. J. *3D Laser-and-Tissue Agnostic Data-Driven Method for Cavity Prediction.*
[Poster Presentation] Fitzpatrick Institute for Photonics Symposium, March 13–14, 2023, Durham, NC, USA.
7. Codd, P. J., Ross, W., Ma, G., Tucker, M., **Prakash, R.**, Raman, A., Zachem, T., Eward, W., Mann, B. *TumorCNC: Engineering an Automated Closed-Loop Robotic System for Neurosurgery.*
[Oral Presentation] Neurosurgical Society of America, June 14, 2022, Maui, HI, USA.
8. Ma, G., **Prakash, R.**, Ross, W., Codd, P. J. *A Data-Driven Method for Robotic Laser Orientation Planning.*
[Poster Presentation] International Symposium on Medical Robotics (ISMR), April 13–15, 2022, Atlanta, GA, USA.

Sensing & Learning

1. Sperber, J., Zachem, T. J., **Prakash, R.**, Chamberlain, G., Cummings, T., Ross, W., Codd, P. J., Goodwin, C. R. *Characterization of the TumorID Technology to Differentiate Tumor from Non-Tumor in Frozen Samples.*
[Oral Presentation] Global Spine Congress, May 31–June 6, 2023, Prague, Czech Republic.

2. Sperber, J., Zachem, T. J., **Prakash, R.**, Chamberlain, G., Cummings, T., Nguyen, A., Hockenberry, H., Ross, W., Codd, P. J., Goodwin, C. R. *Laser-Induced Endogenous Fluorescence Spectroscopy Produces Distinct Spectral Signatures in Pathology-Prepared Tumor Samples*.
[Oral Presentation] 39th AANS/CNS Spine Summit, March 16–19, 2023, Miami, FL, USA.
3. Sperber, J., Zachem, T. J., **Prakash, R.**, Nguyen, A., Hockenberry, H., Owolo, E., Charles, A., Ross, W., Codd, P. J., Goodwin, C. R. *Classification of Fresh Spine Tumor, Muscle, and Bone Using Intraoperative Laser-Induced Endogenous Fluorescence Spectroscopy*.
[Poster Presentation] 39th AANS/CNS Spine Summit, March 16–19, 2023, Miami, FL, USA.
4. Raman, A., Zachem, T., **Prakash, R.**, Park, C., Ma, G., Ross, W., Codd, P. J. *Automated Detection of Sarcoma Tissue in a Murine Model Using a Portable Endogenous Fluorescence Spectroscopy Device*.
[Poster Presentation] International Symposium on Medical Robotics (ISMR), April 13–15, 2022, Atlanta, GA, USA.
5. **Prakash, R.**, Srivastava, A. A., Ni, X. *Methods for Characterization of Mechano-Acoustic Speech Information*.
[Poster Presentation] Duke MEMS Non-Thesis Defense, November 17, 2021, Durham, NC, USA.

Clinical & Healthcare Analytics

1. **Prakash, R.**, Dupre, M. E., Østbye, T., Xu, H. *A Rule-Based Framework to Identify Severity of Dementia from Unstructured Electronic Health Record Data*.
[Poster Presentation] Alzheimer's Association International Conference (AAIC), July 16–20, 2023, Amsterdam, Netherlands.

REVIEWS

IEEE Transactions on Robotics (**TRO**)
IEEE/ASME Transactions on Mechatronics (**TMECH**)
IEEE Robotics and Automation Letters (**RAL**)
Journal of Medical Robotics Research (**JMRR**)
Conference on Robot Learning (**CoRL**)
IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)
IEEE International Conference on Soft Robotics (**RoboSoft**)
International Symposium on Medical Robotics (**ISMR**)
IEEE EMBS – Body Sensor Networks (**BSN**)

MEMBERSHIPS

American Society of Mechanical Engineers (**ASME**)
Institute of Electrical and Electronics Engineers (**IEEE**)

SPONSORED RESEARCH & GRANTS

May 2025 – July 2026

Stryker Corporation (\$346K)

Role: Project Scoping & Strategy; Led Proposal Writing & Assembly; IP Coordination

PI: Patrick Codd, Duke University

Optimal Surgical Usage of Ultrasonic Aspirator for Soft-tissue Resection

Project will identify optimal operating parameters for an ultrasonic aspirator to minimize collateral damage during soft-tissue resection. The work involves developing a robotic platform for evaluating aspirator performance and analyzing operating-room data to understand current clinical usage, including surgeon- and tissue-dependent variations.

May 2025 – May 2026

Bass Connections Program (\$23K)

Role: Team Lead; Project Manager; Instructor of Record

PI: Patrick Codd, Duke University

Reimagining Surgery for Rural Needs: Robotics Teleoperation

The project is developing a teleoperated laser surgery platform to expand access to advanced surgical care in rural clinics. The system will enable precise, minimally invasive soft-tissue procedures requiring rapid hemostasis and resection.

May 2024 – April 2025

Bass Connections Program (\$5K)

Role: Team Lead; Led Proposal Writing

Faculty Mentor: Patrick Codd, Duke University

Enhancing Precision Laser Surgery with Flexible Sensors

Project focused on improving the safety and precision of laser-based surgery by developing improved sensory tools for surgeons.

March 2023 – April 2024

India Initiative at Duke (\$2K)

Role: PI; Led Proposal Writing; Built Clinical Collaborations in India

Faculty Mentor: Patrick Codd, Duke University

Portable Tumor Diagnostic Solutions for Resource Constrained Communities

Project evaluated the adaptability and efficacy of a portable tumor diagnostic sensor for use in resource-constrained clinics in Bihar, India. Work included establishing collaborations with S.S. Hospital and IIT Patna and securing the necessary approvals to conduct feasibility studies.

RESEARCH EXPERIENCE

Aug 2026 – Present

Multimodal Sensing Driven Autonomous Surgical Robots

Mentor: Axel Krieger, Ph.D. (Postdoctoral Advisor, Johns Hopkins University)

Project focuses on developing multimodal perception and sensing frameworks for autonomous surgical robots, integrating imaging, force, acoustic, and other intraoperative signals to enable real-time scene understanding, tissue interaction assessment, and safe adaptive robotic decision-making.

June 2022 – July 2026

Automated Closed-Loop Robotic Laser Ablation of Pathologic Tissues

Mentor: Patrick Codd, M.D., FAANS (Thesis Advisor, Duke University)

Project includes designing and prototyping a closed-loop laser ablation system for targeted soft-tissue removal, integrating machine-learning-based ablation models, image-guided sensing, and laser-tissue interaction studies with dexterous robotic manipulators.

April 2024 – July 2026

Drone Action Control Using Acoustic Signatures

Mentor: Boyuan Chen, Ph.D. (Co-advisor, Duke University)

Project aims to enable a drone to track and follow another using only acoustic input from a custom drone-mounted microphone array. The system allows open-loop drone coordination based on passive acoustic signatures in low-visibility, low-resource environments.

September 2020 – December 2021

Speech-based Physiological State Estimation

Mentor: Xiaoyue Ni, Ph.D. (Master's Advisor, Duke University)

Project investigated the design and fabrication of a multimodal, flexible epidermal sensor patch for transient vital-sign measurements related to neurodegenerative disease assessment.

June 2021 – December 2021

Alzheimer's Disease-Related Dementias (AD/ADRD) Severity Assessment from Unstructured EHR

Mentor: Hanzhang Xu, Ph.D., R.N., FAAN (School of Nursing, Duke University)

Project investigated pathways to predict AD/ADRD severity at the time of diagnosis in underrepresented communities using Duke's unstructured EHR data.

August 2018 – April 2019

Experimental Study of Ledinegg Instability

Mentor: P. Bangaru Babu, Ph.D. (Undergraduate Advisor, National Institute of Technology, Warangal)

Project involved designing and fabricating a robust, leakproof, low-cost open-loop minichannel test setup to study hydrodynamic instabilities related to Ledinegg behavior in microchannel fluid flow.

May 2018 – July 2018

Prototype Development to Study Ledinegg Instability

Mentor: Debjyoti Banerjee, Ph.D. (Texas A&M University)

Project included an initial literature review and prototype development to experimentally study Ledinegg instability and explore real-time control strategies for high-energy component cooling applications.

May 2017 – July 2017

Microchannel Flow Boiling Heat Sink Development

Mentor: Poh Seng Lee, Ph.D. (National University of Singapore)

Project involved simulated and experimental studies of novel hybrid microchannel heat-sink designs using closed-loop flow-boiling experiments.

May 2016 – June 2016

Sensor Suite for Underwater Robot

Mentor: Atul Thakur, Ph.D. (Indian Institute of Technology, Patna)

Project involved investigating depth-measurement options for an underwater mobile robot.

ENTREPRENEURSHIP AND BUSINESS CAREER

Sustainable Living Lab, Singapore

May 2019 – August 2020

Co-Lead, India

Designed and implemented new technology ventures and delivered Intel's global AI curriculum for non-technical audiences AI for Youth. Formulated and led "Futures+," a foresight-driven community innovation program spanning Bhutan, India, Indonesia, and Singapore.

TEACHING & MENTORSHIP

Coursework and Curriculum Development

Instructor of Record

Fall 2025–Spring 2026

GLHLTH 395T / 396T / 796T

Reimagining Surgery for Rural Needs

Bass Connections – Global Health, Duke University

Designed and led an experiential, research-based course that used structured check-ins and reflective sessions to support interdisciplinary teams developing a teleoperated laser-surgery platform for rural clinics. The course covers optomechanical system assembly, basic optics, and integrated hardware–software development informed by an IRB-approved clinical user study.

Course evaluation rating: 4.25/5 (Fall 2025)

Guest Lecturer

Spring 2026

ME/BME 555

Engineering Technology in Biomedical Applications

Department of Mechanical Engineering and Materials Science/ Biomedical Engineering, Duke University

Delivered a dedicated lecture on **Robotics and AI for Laser Surgery**, to highlight my doctoral research work on building intelligent surgical robotic platforms, multimodal signal processing, and machine learning for closed-loop precision soft tissue surgery. Highlights included challenges of intraoperative sensing, such as OCT, and the role of neural state representation methods in overcoming the limitations in speed and query, as well as the development of custom opto-mechanical hardware for imaging and laser resection control.

Guest Lecturer

Spring 2025, Spring 2026

ME 555

Introduction to Medical Robotics and Surgical Technologies

Department of Mechanical Engineering and Materials Science, Duke University

Delivered a dedicated lecture on **Machine Learning for Medical Robotics**, guiding students from diverse backgrounds through common strategies and pitfalls in domains with strict class imbalance and limited data. Developed scaffolded self-learning plans tailored to different levels of ML experience to support equitable and effective skill development.

Curriculum Developer

Summer 2024

Collaborative Expeditions Grant

Introduction to Medical Robotics and Surgical Technologies

Department of Mechanical Engineering and Materials Science, Duke University

Identified teaching bottlenecks and developed scaffolded self-learning modules on **Design of Experiments, Machine Learning, and Literature Review** as part of a Bass Connections Collaborative Expeditions Grant with Prof. Siobhan Oca. The modules integrated reflections and case studies to promote deeper conceptual understanding and open discussion among students with varied backgrounds.

Guest Lecturer

Spring 2024

ME 555 (special topics)

Introduction to Medical Robotics and Surgical Technologies

Department of Mechanical Engineering and Materials Science, Duke University

Delivered a dedicated lecture on **Design of Experiments**, using active learning and problem-focused examples to help students connect concepts such as confounding variables, sources of bias, parameter selection, and appropriate statistical tests to real experimental workflows.

Teaching Assistant

Spring 2021, Fall 2021, Spring 2022

ME 555 (special topics)

Experiment Design & Research Methods

Department of Mechanical Engineering and Materials Science, Duke University

Worked with first-year master's students and senior undergraduates to scope feasible research projects, conduct regular check-ins, and provide technical support in rapid prototyping and research portfolio development within the capstone design sequence.

Mentorship

Mentored high school, undergraduate, and graduate researchers across robotics, sensing, and medical technologies, with contributions leading to conference posters, journal submissions, and prototype developments.

High School Students

1. **Sarah Dias**, Research Triangle High School — *Multimodal Tissue Boundary Test Bench*
 - IEEE ICRA ELSR Poster 2025 & IEEE-EMBS BSN 2025 Paper, IEEE TMRB Submission [Conditionally Accepted]
2. **Prithu Kolar**, NC School of Science and Mathematics — *Acoustic and Video Processing and Real-time Streaming*
 - ICRA ELSR Poster 2025 & IEEE-EMBS BSN 2025 Paper
 - Next:** Park Scholar, NC State University
3. **Felicity Lipchak**, NC School of Science and Mathematics — *Acoustic and Video Processing and Real-time Streaming*
 - **Next:** Dartmouth College

Undergraduate Students

1. **Devi Yuliarti**, Duke University — *Pneumatic Tactile Sensor*
 - (Co-First Author) IEEE RoboSoft 2026; IEEE ICRA 2026
2. **Nikhil Aaryath**, Duke University — *Ultrasonic Aspirator (UA)-Based Tissue Estimation and Optimization*
 - Assisting the robotic-arm-based data-collection pipeline for UA with real-time visual and acoustic monitoring.

3. **Zacharias Chen**, Duke University — *Multimodal Tissue Boundary Test Bench*
 - IEEE TMRB Submission [Conditionally Accepted]
IEEE ICRA ELSR Poster 2025 & IEEE-EMBS BSN 2025 Paper
Next: Medtronic, Hugo Team
4. **Alexa Cahilig**, Duke University — *Multimodal Tissue Boundary Test Bench*
 - IEEE ICRA ELSR Poster 2025 & IEEE-EMBS BSN 2025 Paper
5. **Aakriti**, IIT Bombay — *Characterization of Mechano-Acoustic Speech Information*
 - Contributed to acoustic feature analysis and experimental characterization.
6. **Akshit Srivastava**, IIT Bombay — *Characterization of Mechano-Acoustic Speech Information*
 - Supported data processing and machine learning for mechano-acoustic speech data.

Graduate Students

1. **Tarun Vinodkumar, PhD Student**, Johns Hopkins University — *Autonomous robots for stroke intervention*
 - Ongoing.
2. **Vincent Wang, PhD Candidate**, Duke University — *Safety critical controls framework for precision robotic surgery*
 - Ongoing.
3. **Yongjun Lu**, Duke University — *Optimizing Ultrasonic Aspirator Parameters for Maximal Tissue Resection*
 - Lead clinical-side data acquisition and processing for ultrasonic aspirator evaluation in the operating room.
Next: Ethicon, J&J Company
4. **Antony Farrand**, Duke University — *Intelligent Acoustic Sensing for Laser-Guided Surgery*
 - Poster presented at the Duke MEMS Master's Research Poster Symposium.
5. **Caroline Chen**, Duke University — *Segmentation & Planning Web Application for Robotic Laser Surgery*
 - IEEE ISMR Poster 2024
Next: Machine Learning Engineer, Google
6. **Changxin Yu**, Duke University — *Unsupervised Learning-Based OCT Segmentation*
 - 3rd Prize, Duke MEMS Master's Research Poster Symposium
Next: Microsoft

SERVICE AND LEADERSHIP

Founding Member Engineering Graduate Ambassadors <i>Duke University</i>	May 2023–Jun 2025
Lab Manager Brain Tool Lab <i>Duke University</i>	Jul 2023–Jul 2026
President Graduate Student Committee <i>Department of Mechanical Engineering & Materials Science, Duke University</i>	Jan 2022–May 2023
Secretariat Member Graduate and Professional Student Government (GPSG) <i>Duke University</i>	Aug 2021–Oct 2022
MEMS Representative Engineering Graduate Student Committee	Aug 2021–May 2022

Duke University

Founder and Mentor

TEDxNITW

National Institute of Technology Warangal

Feb 2017–May 2019

Facilitator & Technical Lead

Innovation Garage (Incubation Center and Makerspace)

National Institute of Technology Warangal

Mar 2016–May 2019

REFERENCES

Axel Krieger, Ph.D.

Associate Professor & Carol Croft Linde Faculty Scholar
Department: Mechanical Engineering
Johns Hopkins University
Email: axel@jhu.edu

Patrick J. Codd, M.D., FAANS

Associate Professor
Departments: Neurosurgery; Mechanical Engineering and Materials Science; Head and Neck Surgery
Duke University
Email: patrick.codd@duke.edu

Boyuan Chen, Ph.D.

Dickinson Family Assistant Professor
Departments: Mechanical Engineering and Materials Science; Electrical and Computer Engineering; Computer Science
Duke University
Email: boyuan.chen@duke.edu

Jerome P. Lynch, Ph.D.

Vinik Dean of Engineering; Fitzpatrick Family University Distinguished Professor of Engineering
Departments: Civil and Environmental Engineering; Electrical and Computer Engineering
Duke University
Email: jerome.lynch@duke.edu

Leila J. Bridgeman, Ph.D.

Assistant Professor
Department: Mechanical Engineering and Materials Science
Duke University
Email: leila.bridgeman@duke.edu

Gerald A. Grant, M.D., MBA, FAANS

Chair of Neurosurgery; Allan H. Friedman Distinguished Professor of Neurosurgery;
Departments: Neurosurgery; Pediatrics; Neurology; Biomedical Engineering
Duke University
Email: gerald.grant@duke.edu